GLA-STDeepLab: SAR Enhancing Glacier and Ice Shelf Front Detection Using Swin-TransDeepLab With Global–Local Attention

Qi Zhu, Huadong Guo, *Member, IEEE*, Lu Zhang, Dong Liang, Zherong Wu, *Member, IEEE*, Yiming Liu, and Zhuoran Lv

***Abstract*— The understanding of glacier and ice shelf front changes is of paramount importance in analyzing their material balance and their significant contributions to global sea-level variations. Recently, the increasing availability of remote sensing images has provided ample data support for calving front detection. By combining massive amounts of data, deep learning techniques have demonstrated great potential in front extraction, as they can automatically generate task-specific features. In this study, in contrast to the traditional framework based on a pure convolutional neural network (CNN) architecture, we propose a novel calving front detection network, global–local attention Swin-transformer-based DeepLabv3+ (GLA-STDeepLab), by incorporating the Swin-transformer module into DeepLabv3+ to enhance the modeling capability of long-range contextual dependencies. We also introduce global–local attention (GLA) mechanisms into the module, striving to foster interactions between the corresponding coarse and fine-grained feature maps. Through extensive experimentation on a diverse and challenging synthetic aperture radar (SAR) database, CaFFe, our network is shown to surpass the current state-of-the-art methods, achieving results with an intersection over union (IoU) of 0.94 and a mean distance error (MDE) of 473 ± 34 m. More meaningfully, we extracted the time-series front information of the Amery Ice Shelf (AIS) from 2015 to 2023 and precisely captured the calving events of the D-28 and D-32 icebergs. The findings demonstrate that our method enables precise monitoring of the dynamic information of glacier and ice shelf fronts, as well as their far-reaching implications for ice sheet mass balance and global climate change. The code used in this study is provided at https://github.com/Tangyu35/Calving-front-detection.**

***Index Terms*— Front detection, global–local attention Swin-transformer-based DeepLabv3+ (GLA-STDeepLab), Swin-transformer, synthetic aperture radar (SAR).**

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Qi Zhu, Huadong Guo, Lu Zhang, Dong Liang, and Zhuoran Lv are with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, also with the International Research Center of Big Data for Sustainable Development Goals, Beijing 100094, China, and also with the College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: qizhu@cbas.ac.cn; hdguo@radi.ac.cn; zhanglu@radi.ac.cn; liangdong@radi.ac.cn; lvzhuoran21@mails.ucas.ac.cn).

Zherong Wu is with the Institute of Space and Earth Information Science, The Chinese University of Hong Kong, Hong Kong, China (e-mail: zherong.wu@link.cuhk.edu.hk).

Yiming Liu is with the School of Earth and Space Sciences, Peking University, Beijing 100871, China, and also with the International Research Center of Big Data for Sustainable Development Goals, Beijing 100094, China (e-mail: yimingliu@pku.edu.cn).

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I. Introduction

OVER the past two decades, there has been an increase in the frequency of retreat and fracturing of glaciers and ice shelf fronts [1]. The disintegration of glaciers and ice shelf fronts is a crucial factor affecting their mass balance [2] and contributing to the global sea level [3]. Changes in calving fronts are influenced by a range of complex factors, including geological structures [4], ocean circulation [5], glacier dynamics [6], meltwater runoff [7], and climate change [8]. However, the sensitivity of calving front changes to these factors and the geographical environment is not yet well understood [9]. As such, there is an urgent need for a temporally and spatially comprehensive database on calving fronts to improve our understanding of the changes in glaciers and ice shelf fronts and their impact on global climate change.

The number of remote sensing images in the polar regions, including optical and synthetic aperture radar (SAR) images, has significantly increased during the past two decades [10]. Although the increasing number of remote sensing images provides ample data resources for extracting glacier and ice shelf fronts, many studies still rely on manual delineation to obtain the ice shelf front, which is labor-intensive and time-consuming [11]. Automated front mapping is therefore a better option due to its lower computational consumption and human labor; it primarily includes feature extraction and deep learning methods [12]. The former uses handcrafted features (the Roberts operator [13], the Sobel operator and brightness profiling [14], and the Canny operator [15], to name only a few) to obtain dynamic changes in fronts. Specifically, multipolarization C-band SAR imagery is utilized to devise an automated detection methodology based on a global-threshold constant false alarm rate technique, enabling the precise extraction of the ice front of the d'Iberville glacier [16]. Furthermore, Hesari et al. [17] extended the analysis by leveraging time-series SAR imagery to investigate the coherent motion and surface velocity of the Drygalski ice tongue These methods have enabled dynamic monitoring of calving fronts in specific regions. However, they rely on handcrafted features and require dense prior knowledge of specific glaciers or ice shelves, making it challenging to transfer them to other areas with different topographies and environments.

Deep learning has been utilized for automatic calving front detection [1], [12], [18], [19], [20], [21] as convolutional neural networks (CNNs) can automatically generate features

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tailored to computer vision (CV) tasks, making them a superior choice for CV applications. The lack of a publicly available calving front dataset has led most methods to use the modified U-Net network model [12], [18], [19], [20], which has been trained on small samples and shown to achieve excellent results [22]. These models typically have minimal or no additional information interaction, apart from skip connections. In addition, deep learning networks based on the CNN framework are limited by the convolutional locality and translational invariance of the CNN and cannot capture global semantic information, significantly restricting the accurate delineation of calving front changes [23].

To address these challenges, we incorporate the transformer module [24], renowned in the natural language processing (NLP) field, into the advanced encoder–decoder model DeepLabv3+ [25]. This solution overcomes the CNN model's limitation in capturing long-range spatial dependencies. Furthermore, we introduce global–local attention (GLA) blocks, which are primarily used to merge global semantic information (obtained from the transformer) and local semantic information (obtained from the CNN), enabling a more precise delineation of the calving front. We demonstrate the effectiveness of our proposed model, GLA-STDeepLab, through extensive experiments on the diverse calving front of the benchmark dataset CaFFe [26], which includes a range of SAR images covering the regions of Antarctica, Greenland, and Alaska.

II. Related Work

A. CNN-Based Calving Front Segmentation Methods

The methods for extracting glaciers and ice shelf fronts presented in the literature are mostly based on enhancements to U-Net, which consists of an encoder structure for extracting semantic information and a symmetric decoder layer for restoring image resolution and achieving precise pixel-level predictions. For instance, Mohajerani et al. [27] employed a modified U-Net to distinguish between calving fronts and background using low-resolution Landsat images. Similarly, Baumhoer et al. [18] utilized an improved U-Net in combination with Sentinel-1 SAR images and the Antarctic TanDEM-X DEM to trace glacier fronts. This method was later employed to analyze the retreat of ice shelf fronts in the circum-Antarctic region over the past two decades [1]. Furthermore, Heidler et al. [20] combined a segmentation network (U-Net) to classify land and water/ice, with an edge detection framework, inspired by holistically-nested edge detection (HED) to precisely map the ice shelf front. This approach was subsequently utilized by Baumhoer et al. [28] to automatically generate the precise locations of 19 400 calving fronts and establish the "IceLines" database, which records the dynamic changes in ice shelf fronts in the Antarctic region.

The U-Net architecture has been replaced by a more cutting-edge approach DeepLabv3+, which leverages dilated convolutions and atrous spatial pyramid pooling (ASPP) layers, as they have been shown to capture multiscale semantic information precisely in recent studies. Zhang et al. [12] integrated optical and SAR imagery for land–ocean discrimination and compared the results with those of U-Net-based methods, finding that DRN-DeepLabv3+ achieved the lowest testing error. Cheng et al.'s work [29], CALFIN, utilized Xception [30] as the backbone, similar to the original DeepLabv3+ network, and generated two prediction masks for land or ocean and coastline or background. Recently, Holzmann et al. [31] employed attention gates in the U-Net architecture to improve segmentation accuracy.

More recently, Gourmelon et al. [26] proposed the CaFFe database for evaluating the ability of deep learning models in front edge detection and validated the diversity and wide application prospects of the dataset using an improved U-Net. Wu et al. [21] introduced the attention-multihooking-deep-supervision method HookNet (AMDHookNet) based on CaFFe, which achieved higher accuracy in calving front detection compared to the results in [26].

Although CNN-based front detection models have shown some promise, extracting accurate front information from the complex scenes of glaciers and ice shelves remains a significant challenge since it is difficult to distinguish between sea ice and land ice, as well as between surface meltwater and ocean, which cannot be achieved solely through local information. Local features may not fully capture the global contextual information, leading to inaccuracies and ambiguities in classification. Thus, an integrative approach that combines both local and global information may be necessary.

B. Global Contextual Information Modeling

Various methods have aimed to integrate global semantic information with local information through attention mechanisms to address the CNN's limitations in attending only to local information. The transformer network model [32] has achieved remarkable success in NLP and inspired the application of attention mechanisms in CV. Woo et al. [33] proposed a plug-and-play convolutional block attention module (CBAM) that enhances network accuracy by acquiring channel and spatial attention. Wang et al. [34] introduced a nonlocal attention model to capture global contextual information between pixels in images, yielding remarkable results.

Remote sensing images are often characterized by complex scenes and rich information, including multiband information in hyperspectral images and polarimetric information in SAR images. Such complexity poses a significant challenge for accurate classification. Attention mechanisms have demonstrated their potential in enhancing network representation and identifying significant regions of an image through weighted assignment. For example, Wang et al. [35] proposed the attention recurrent convolutional network (ARCNet) for very high-resolution (VHR) remote sensing image classification, which outperformed advanced methods in terms of accuracy. To address object detection in VHR images, Alhichri et al. [36] developed a multiscale visual attention network (MS-VAN) with skip connections and multiscale attention. In addition, Tang et al. [37] designed the attention consistent network (ACNet), which combines global semantic information and local information for scene classification. These innovative approaches demonstrate the potential of attention mechanisms in improving the accuracy and efficiency of remote sensing image analysis.

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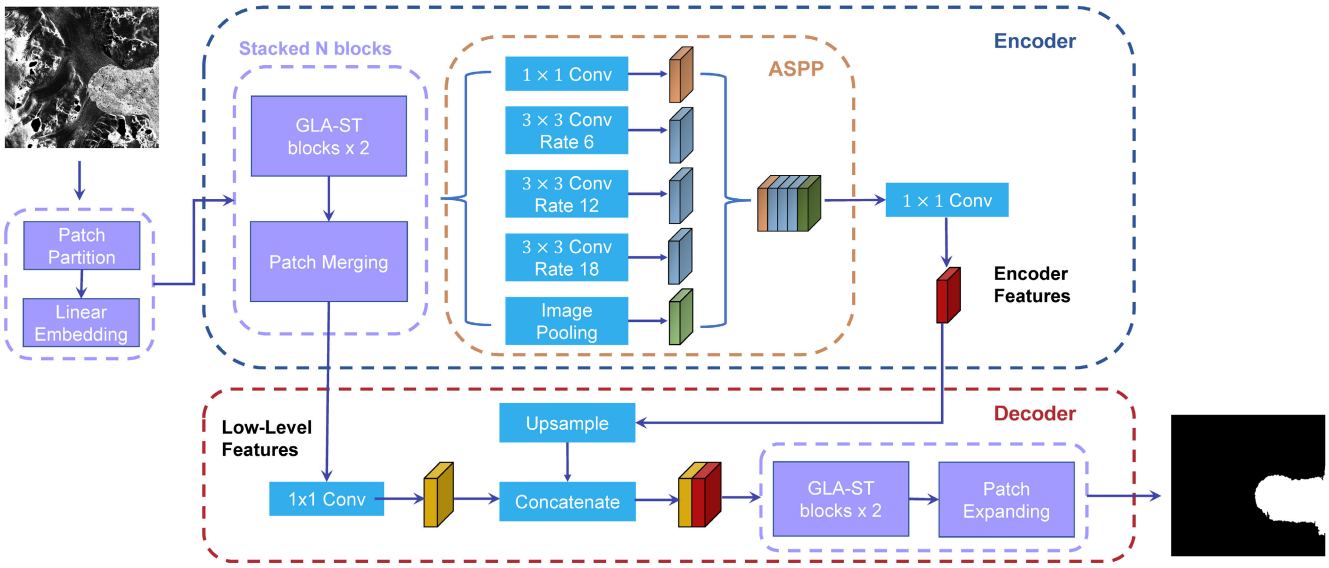


Fig. 1. Architecture of GLA-STDeepLab, which extends the encoder–decoder structure of DeepLabv3+ with GLA-STs. For the architecture, the encoder block is shown in the light blue dotted box, and the decoder block is in the brown-red. The input and the output are the SAR imagery and the segmentation result, respectively.

C. Transformer-Based Semantic Segmentation Methods

The CNN model is imbued with the inductive bias of convolutional locality and translational invariance, which makes it easier to train, with fewer parameters and higher efficiency. Nonetheless, this inductive bias also leads the model to disregard long-range spatial dependencies, thereby impeding further improvements [38]. Recently, transformer has made remarkable strides in the NLP field, leveraging its potent spatial modeling ability in the context of long-range contextual dependencies. This achievement has captured the attention of the CV community. In recent years, transformer has demonstrated superior performance across diverse CV tasks [24], [39], [40], [41], [42], [43], [44]. In light of these developments, a growing number of researchers have started to adopt the transformer approach to address challenges such as remote sensing image classification [45], [46], object detection, and change detection [47], [48], [49], with a particular emphasis on semantic segmentation [23], [50].

Currently, transformer-based semantic segmentation methods still adhere to the encoder–decoder layer structure. These methods can be broadly classified into two categories based on different combinations of encoder–decoder layers [23]. The first category comprises the pure transformer structure, which constructs a semantic segmentation network through an encoder–decoder structure with transformer blocks instead of a traditional convolutional layer. The pyramid vision transformer (PVT) [42] was the first hierarchical transformer structure, proposing a progressive shrinking pyramid and spatial reduction attention. SegFormer [43] builds upon the PVT design philosophy and adopts a simple yet efficient transformer-based segmentation model that uses a multilayer perceptron (MLP) as the encoder. CrossFormer [39], which also features a hierarchical pyramid structure, introduces a cross-scale embedding module, long-short distance attention, and dynamic bias to capture global–local information.

The second category of methods generally adopts a hybrid structure, usually consisting of a transformer-based encoder and a CNN-based decoder. Convolution-enhanced image transformers [51] apply convolutional layers to the image-token module to obtain low-level feature information. LocalViT [52] introduces depthwise convolutions (DWConvs) into the local feature extraction of the ViT model. DC-Swin [53] employs the Swin-transformer as the encoder layer and designs a CNN-based decoder with dense connections for VHR image segmentation. Cao et al. [54] introduced hierarchical Swin-transformer modules into the classic semantic segmentation network U-Net, yielding Swin-Unet, which outperforms many pure CNN methods. Similarly, Azad et al. [55] incorporated Swin-transformer modules into DeepLabv3+, achieving excellent accuracy while significantly reducing model complexity.

In this study, we adopt a hybrid structure, combining the transformer module with the DeepLabv3+ architecture, to fully leverage the strengths of the transformer and CNN modules. Furthermore, a global–local attention block, aiming to merge global–local information, is introduced into our model, namely, global–local attention Swin-transformer-based DeepLabv3+ (GLA-STDeepLab).

III. Methodology

In this study, we present a novel network architecture, GLA-STDeepLab (shown in Fig. 1), designed for the detection of glacier and ice shelf fronts. Unlike the traditional detection framework based on pure CNNs [12], [18], [20], [21], we incorporate the Swin-transformer module into the DeepLabv3+ and introduce GLA mechanisms into the Swin-TransDeepLab module, aiming to integrate the advantages of the traditional CNN and transformer structures to enhance the expressive ability of the network. A more detailed account of the constituent elements is provided as follows.

Transformer-based DeepLabv3+ (GLA-STDeepLab) is shown in Fig. 1. The GLA-STDeepLab is primarily comprised

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of global–local attention Swin-transformer blocks (GLA-STs), as well as a DeepLabv3+ encoder–decoder structure. A more detailed account of the constituent elements is furnished in the following.

A. Swin-TransDeepLabv3+ Model

DeepLabv3+ is a cutting-edge semantic segmentation network model that combines the encoder–decoder structure with atrous convolution (or dilated convolution) [25]. It has shown great success in various remote sensing segmentation tasks [56], [57], [58], especially in the field of front detection [12], [29], [59], [60]. Recently, transformer-based CNN models have gained considerable attention due to their ability to model long-range contextual dependencies, which is significantly different from that of traditional CNNs, and because of the low computational burden of Swin-transformer blocks compared to that of the Vision Transformer [24]. In this regard, Azad et al. [55] proposed the transformer-based DeepLabv3+ (TransDeepLab) model and achieved exceptional performance. Here, we present a succinct overview of TransDeepLab and focus on the global–local attention Swin-transformer blocks that are proposed in this study. For more detailed information on TransDeepLab, the authors recommend [55].

1) Encoder: TransDeepLab employs stacked Swin-transformer blocks in the encoder (the light blue dotted box in Fig. 1). Initially, the input token ($H \times W \times C$) is partitioned into patches to produce a $(H/4) \times (W/4) \times 16C$ tensor, which is then processed by two successive Swin-transformer blocks. Notably, in our experiments, GLA-STs, which will be discussed in Section III-A, are adopted instead of Swin-transformer blocks. Following this, a sequence of stacked GLA-ST blocks is utilized to gradually decrease the image resolution while increasing its dimension (similar to CNN structures). The resulting mid-level feature map is then forwarded to the ASPP layer, enabling the capture of multiscale semantic information.

2) Atrous Spatial Pyramid Pooling: The spatial resolution of the deep feature maps is significantly reduced after passing through a series of GLA-STs and patch merging operations in the encoder. To compensate for this loss in spatial representation and generate a multiscale representation, the DeepLabv3+ model employs the ASPP module. It is worth noting that in TransDeepLab, swin spatial pyramid pooling (SSPP) is used instead of ASPP to extract both global and local contextual information [55]. However, in this study, the more efficient GLA-ST layers are used to obtain global semantic information, and thus, the ASPP layer is still utilized. The output features are concatenated after passing through several groups of atrous convolutions with different atrous rates (1, 6, 12, and 18 in GLA-STDeepLab), and the encoder layer's output is obtained through a 1×1 convolutional layer.

3) Decoder: In the decoder layer (the brown-red dotted box in Fig. 1), the features from the encoder undergo an upsampling process, increasing their resolution by a factor of 4. Simultaneously, these upsampled features are concatenated with low-level features, which can effectively address the loss of detail due to downsampling [25]. To restore the original image resolution of $H \times W$, a series of GLA-STs accompanied by path-expanding operations are employed.

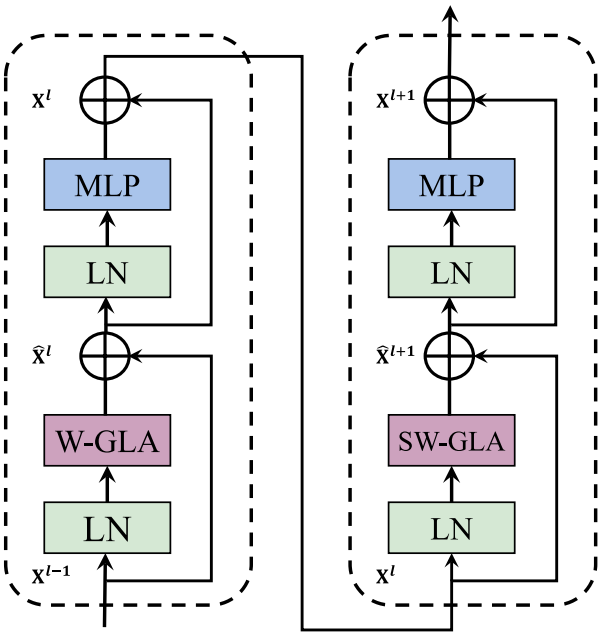


Fig. 2. Architecture of GLA Swin-transformer blocks, which employs GLA modules instead of MSA modules. This architecture is adapted from [61].

B. Global–Local Attention Swin-Transformer Blocks

Fig. 2 shows the global–local attention Swin-transformer blocks, an advanced version of Swin-transformer blocks, which incorporate global–local attention into transformer. The original Swin-transformer blocks include two concatenated sets of self-attention mechanisms with shifted windows that efficiently capture global spatial dependencies [61]. However, while the global context is crucial for calving front detection, local information is still valuable for preserving rich spatial details [23].

Inspired by [23], our global–local attention blocks (Fig. 3) consist of two branches that separately detect global and local contextual information.

The global branch (light green box in Fig. 3) employs W-MSA to capture global semantic information. We utilize a window partition operation that transforms the data into a 1-D sequence. This sequence is subsequently divided into query (Q), key (K), and value (V) vectors [32]. After introducing the relative position bias, global contextual information is calculated according to the self-attention equation

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \tag{1}$$

The local branch (blue box in Fig. 3) utilizes DWConvs in three parallel convolutional layers with kernel sizes of 1, 3, and 7 to extract local information. The choice of DWConv over traditional convolution was driven by the need to conserve computational resources, especially given the significant computing power required for training transformer models [24], [61]. Subsequently, the outputs of the three convolutional layers are passed through a batch normalization layer and

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combined through elementwise summation to obtain local contextual information.

In the last step, the global and the local contextual information extracted from the feature map are combined through elementwise summation. The result is then processed through a 3×3 convolutional layer followed by a batch normalization layer. A 1×1 convolutional layer is used to adjust the number of channels to obtain the GLA.

C. Loss Function

We optimize the GLA Swin-transformer blocks using an auxiliary optimizer in addition to calculating the primary loss (L_p) of the network predictions in model training [62], [63]. The primary loss function is a combination of Dice loss [see (2)] and binary cross-entropy (BCE) loss [see (3)], where the Dice loss (L_{Dice}) can effectively handle cases of sample imbalance [64], while the BCE loss (L_{BCE}) is a traditional method for computing binary classification loss in machine learning [65]. The primary loss is calculated by (4)

$$L_{\text{Dice}}(y, \hat{y}) = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i} \tag{2}$$

$$L_{\text{BCE}}(y, \hat{y}) = -\frac{1}{N}\sum_{i=1}^{N}\left(y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right) \tag{3}$$

$$L_p = \alpha \cdot L_{\text{Dice}} + \beta \cdot L_{\text{BCE}} \tag{4}$$

where y represents the label of the sample (0 or 1 in the experiment), y_{hat} is the predicted probability of the network, and α and β are set to 1 and 0.5, respectively.

The auxiliary loss (L_{aux}) function comprises only BCE loss, which uses the feature maps outputted from the GLA-STs to calculate the loss. To better integrate with the primary loss function, the value of the auxiliary loss function is multiplied by a factor c. Therefore, the loss can be represented as follows:

$$L = L_p + \gamma \cdot L_{\text{aux}} \tag{5}$$

where γ is set to 0.5, consistent with [23].

IV. Experiments

A. Datasets

Our experiments were carried out using the CaFFe dataset [26], a benchmark dataset for SAR imagery calving front detection. The CaFFe dataset encompasses seven tidewater glaciers, chosen as the foundation for training and assessing automated techniques aimed at mapping calving fronts in SAR imagery. These glaciers include the Dinsmoor–Bombardier–Edgeworth (DBE) and Sjoren Inlet (SI) systems situated on the Antarctic Peninsula. In addition, the dataset features the Crane, Mapple, and Jorum Glaciers within the Larsen B embayment, along with the Jakobshavn Glacier (JAC) and Sermeq Kujalleq in Greenland, as well as the Columbia Glacier in Alaska, collectively forming a diverse selection for comprehensive analysis and method development. The dataset includes highly detailed manually traced glacier fronts and their corresponding original SAR imagery, which were obtained from six distinct SAR satellites, encompassing the timeframe of 1995–2020, with diverse spatial resolutions and polarizations. The specifications and parameters of the SAR sensors and imagery are provided in Table I. In the CaFFe database, the SAR imagery from the Columbia and Mapple glaciers is used as the test set, and all the rest of the data are used as the training set. The high diversity and complexity present a considerable challenge for front detection using the CaFFe dataset [21].

TABLE I
SUMMARY FOR THE SAR SATELLITES AND SPECIFICATIONS OF THE CAFFE DATASET. THIS TABLE IS ADAPTED FROM [26]

Platform	band	Time interval	Ground range resolution (m)
ERS-1/2	C band	13 November 1995–2 April 2010	20
RADARSAT 1	C band	10 September 2000–20 January 2008	20
Envisat	C band	5 December 2003–3 July 2010	20
ALOS	L band	18 May 2006–3 March 2011	16.7
TerraSAR-X and TanDEM-X	X band	13 October 2008–20 May 2016	6.7
Sentinel-1 A/B	C band	18 December 2015–12 June 2020	20

The CaFFe dataset comprises preprocessed SAR images as PNG files for network training and two sets of corresponding labels [26]. The initial set of labels focuses on the binary segmentation model, specifically targeting "front" labels. In contrast, the second set includes a multiclass segmentation model featuring "zones" labels. These labels comprise various classes, namely, the "ocean and ice-melange," "rock outcrops," "glacier," and "no-information-available" classes (NA-Area). As this study focuses on glacier and ice shelf front extraction, we merged the "ice-melange," "rock outcrops," and "NA-Area" classes into a "nonglacier" class for subsequent network training. In addition, the substantial size of the CaFFe dataset (e.g., 4476×2107 or 2639×3143) precludes the possibility of directly feeding it into networks. Thus, we cropped the images to a size of 384×384, consistent with the Swin-transformer-Large (Swin-L) model in [61] while ensuring a 50% overlap between any two images to restore edge regions. Finally, in our experiments, the training set included 8200 images and the test set included 2600 images.

B. Evaluation Metrics

This experiment used two main categories of evaluation metrics. The first category consisted of five common metrics: pixel accuracy, mean intersection over union (MIoU), precision, recall, and F1 score [66]. The equations for calculating these metrics are listed as follows:

$$\text{PixelAccuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{6}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{TN}} \tag{7}$$

$$\text{MIoU} = \frac{1}{n}\sum_{i=1}^{n} \text{IoU} \tag{8}$$

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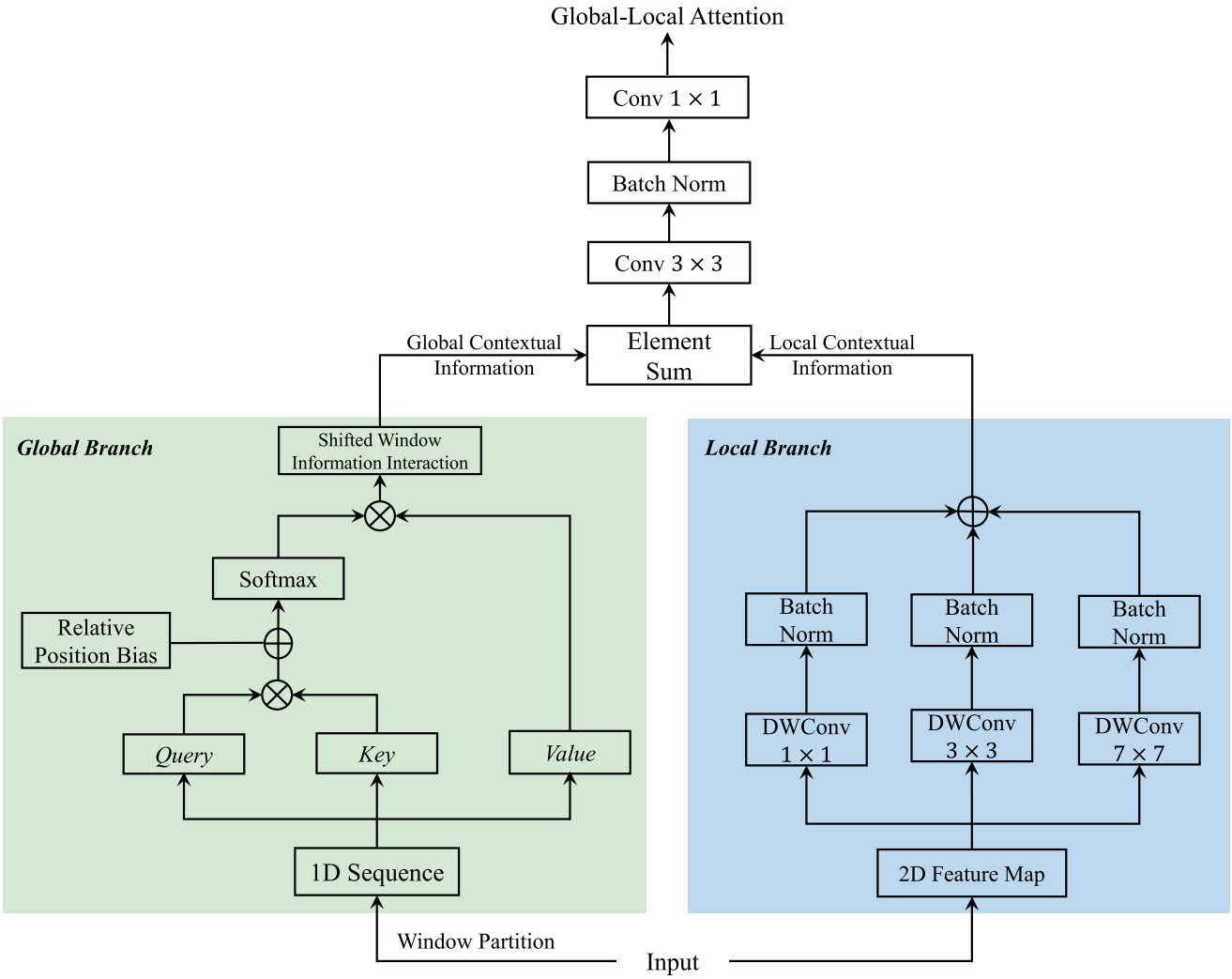


Fig. 3. Structure of global–local attention.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{9}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{10}$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{11}$$

where true positives (TPs) and false positives (FPs) represent cases where the model predicts the positive class as positive and negative, respectively; and true negatives (TNs) and false negatives (FNs) represent cases in which the model predicts the negative class as negative and positive, respectively [66].

The mean distance error (MDE) is the second evaluation metric category used in our experiments, which is widely employed in front detection [12], [18], [20], [26]. MDE measures the mean distance between the model prediction and the nearest pixel in the ground truth, thereby evaluating the performance of the different models. For a detailed description of MDE, the authors recommend [26].

C. Implementation Details

The models were implemented using the PyTorch deep learning framework on a single RTX8000 GPU. Consistent with [55], we chose stochastic gradient descent (SGD) as the optimizer and trained each model for 150 epochs. The learning rate was set to 0.05, and an adaptive decay value of $1e^{-4}$ was used. The largest batch size was set to 64 to fully utilize the GPU's power.

V. Results and Discussion

A. Quantitative and Qualitative Evaluation

Table II shows the benchmark evaluation metrics of the GLA-STDeepLab method. Our algorithm demonstrated outstanding performance in the segmentation of ice shelves and glaciers. Our study focused on the delineation of imagery into two distinct categories, ice shelves or glaciers, and others, as opposed to the four categories utilized in the benchmark literature [26]. Specifically, the pixel accuracy reached an impressive 0.9676, with an F1 score of 0.9541 and an intersection over union (IoU) of 0.9130. These metrics underscore the high accuracy and low misclassification rate of our algorithm in correctly classifying glacier pixels while maintaining a

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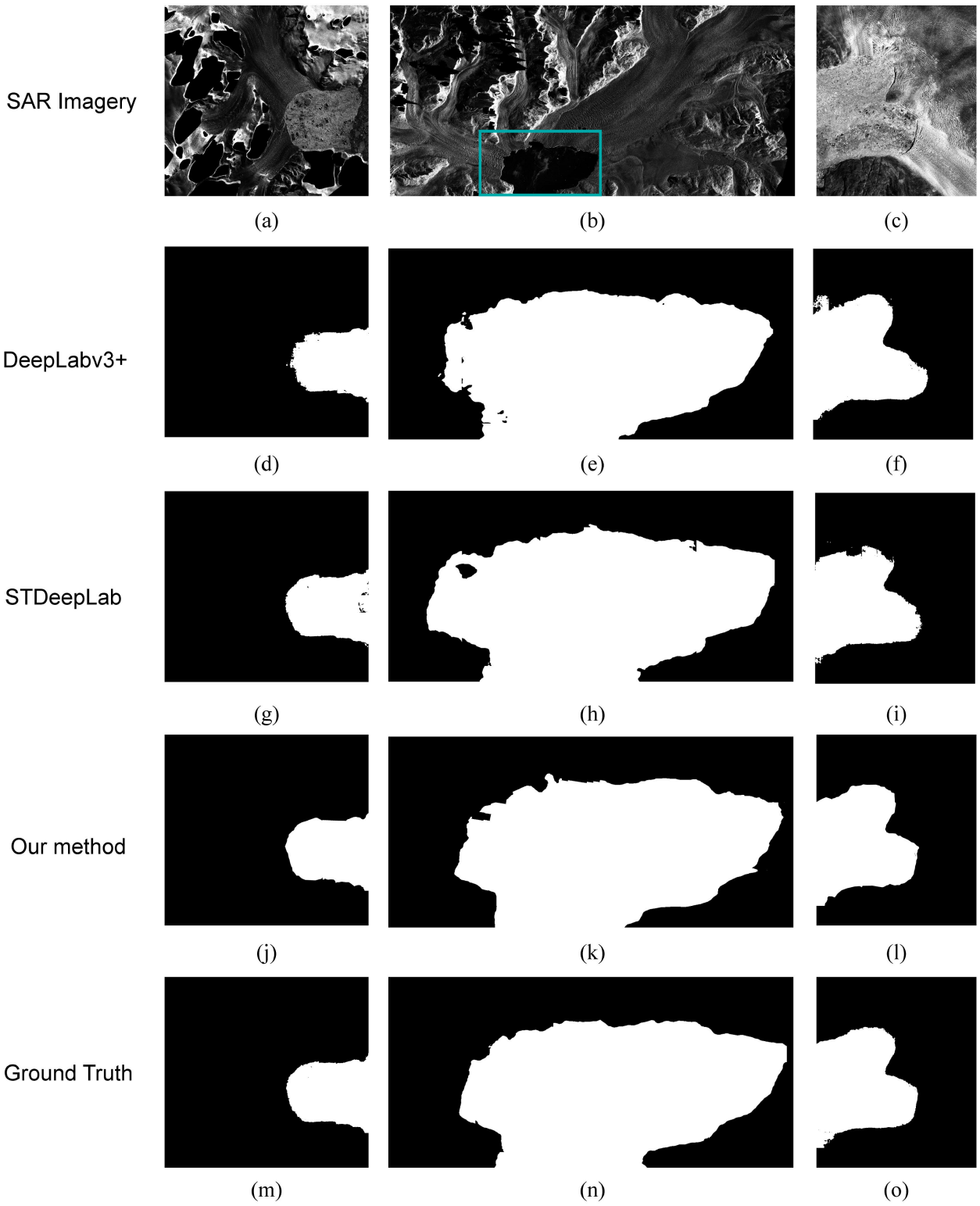


Fig. 4. Qualitative comparison of ground truth, GLA-STDeepLab segmentation results, STDeepLab segmentation results, and corresponding original SAR images. For the segmentation maps, white represents the ice shelves and glaciers, and black represents the others. (a) Image of the DBE Glacier acquired on April 5, 2012 by the TSX satellite. (b) Image of the Columbia Glacier acquired on September 3, 2011 by the TDX satellite. For visualization, only the Columbia Glacier area, blue box in (b) is displayed. (c) Image of Jacobshavn Glacier acquired on December 21, 2011 by the TSX satellite. (d)–(l) are the glaciers and ice shelves segmentation results of DeepLabv3+, STDeepLab and our method of image (a)–(c), respectively. (m)–(o) are the ground truth of image (a)–(c).

commendable balance between precision and recall. We show the ground truth and the segmentation maps generated by GLA-STDeepLab, as well as STDeepLab without global–local attention (as discussed in detail in Section V-B) in Fig. 4.

The delineation of ice shelves is carried out following segmentation, using the MDE as the corresponding evaluation metric. Three accurate predictions and three incorrect predictions of calving fronts using the GLA-STDeepLab

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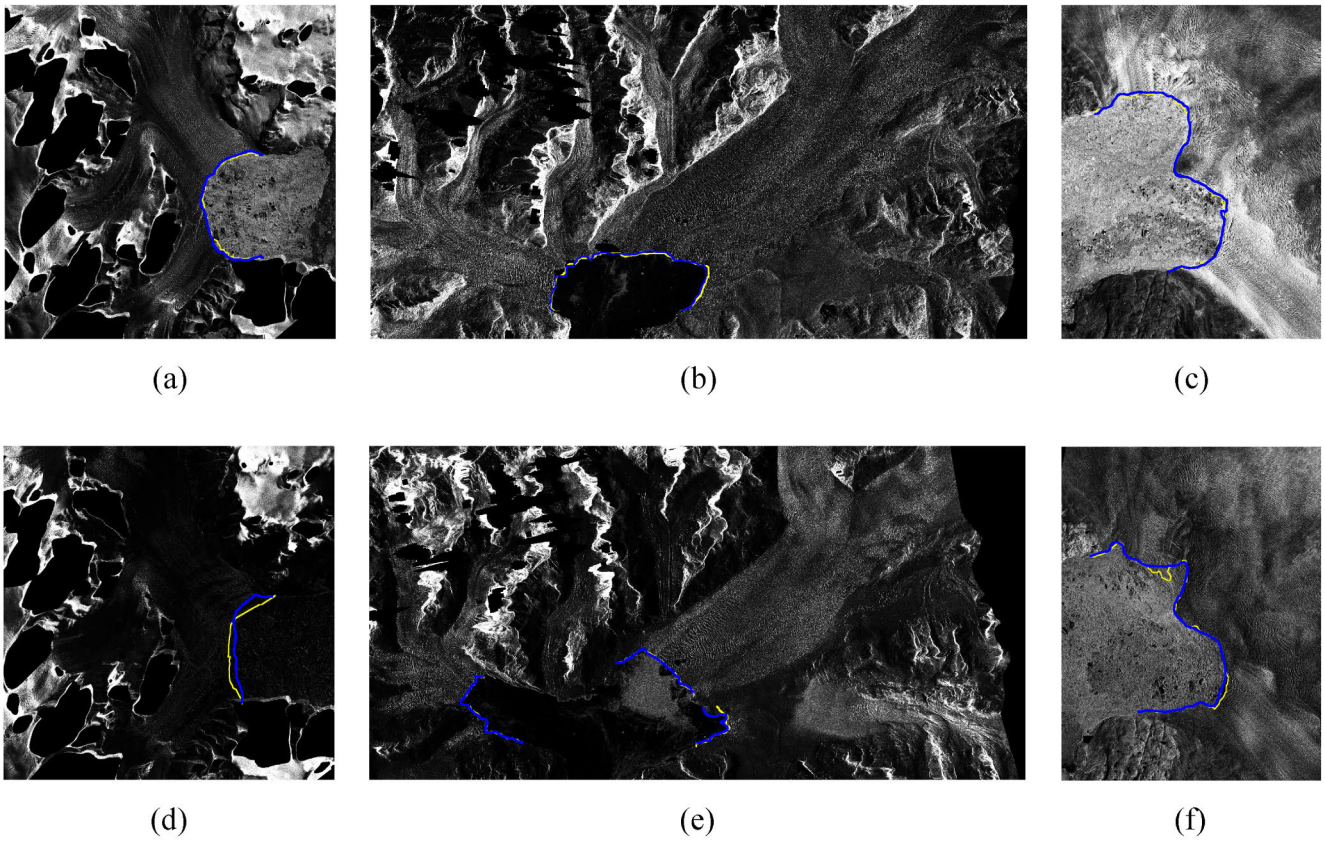


Fig. 5. Visualization of the Glacier front detection of the GLA-STDeepLab. (a) Image of the DBE Glacier acquired on April 5, 2012 by the TSX satellite. (b) Image of the Columbia Glacier acquired on September 3, 2011 by the TDX satellite. For visualization, only the Columbia Glacier area, blue box in (b) is displayed. (c) Image of Jacobshavn Glacier acquired on December 21, 2011 by the TSX satellite. (d) Image of the DBE Glacier acquired on January 1, 2014 by the TSX satellite. (e) Image of the Columbia Glacier acquired on February 11, 2016 by the TDX satellite. (f) Image of Jacobshavn Glacier acquired on September 6, 2009 by the TSX satellite. The yellow and blue lines represent the ground truth and the prediction, respectively.

TABLE II
EVALUATION METRICS (%) FOR THE GLACIER AND ICE SHELVES SEGMENTATION ON THE TEST SET

Scope	Pixel Accuracy	Precision	Recall	F_1	IoU
Glacier	96.76	95.46	95.36	95.41	91.30
Others	98.62	98.68	98.76	98.72	97.40
Mean	97.69	97.07	97.06	97.07	94.35

TABLE III
COMPARISON OF MDE (M) BETWEEN BASELINE AND GLA-STDEEPLAB BASED ON SEASONS ON THE TEST SET

Method	Glacier	MDE	Summer-MDE	Winter-MDE
Baseline [26]	All	887±189	738±111	1054±308
	Columbia	1032±227	907±131	1157±350
	Mapple	150±24	140±26	173±33
Our method	All	602±76	568±61	864±92
	Columbia	873±134	780±92	987±199
	Mapple	70±14	62±9	76±17

method are shown in Fig. 5, where the true glacier front is depicted in yellow, while the glacier front delineated by our method is represented in blue. The primary error arises from the minimal contrast and tone variations observed in certain SAR images [see Fig. 5(d) and (e)], which poses a significant challenge in glacier extraction. Furthermore, the presence of floating ice presents another major obstacle. As a substantial amount of summer sea ice emerges, the precision of extracting the glacier front [see Fig. 5(f)] notably diminishes. Table III presents the MDE values of the GLA-STDeepLab and baseline methods for all glaciers and ice shelves in different seasons. Notably, GLA-STDeepLab achieved an MDE of 453 ± 34 m across the entire test set, representing a significant 46.7% improvement over the baseline, indicating the enhanced accuracy of the extracted front positions.

Moreover, our approach yielded MDE values of 368 ± 42 and 573 ± 64 m for summer and winter, respectively, exhibiting an impressive 50.1% and 45.6% improvement over the baseline. These findings show the efficacy and high precision of the GLA-STDeepLab method. The model excelled in obtaining more precise results for summer imagery, primarily due to the absence of sea ice coverage in the vicinity of glaciers and ice shelves, thereby minimizing model errors. Conversely, the ice mélange in winter near the sea ice and glacier terminus resulted in similar backscatter coefficients to those of the glaciers, leading to a significant reduction in detection accuracy. For the Columbia Glacier, precise

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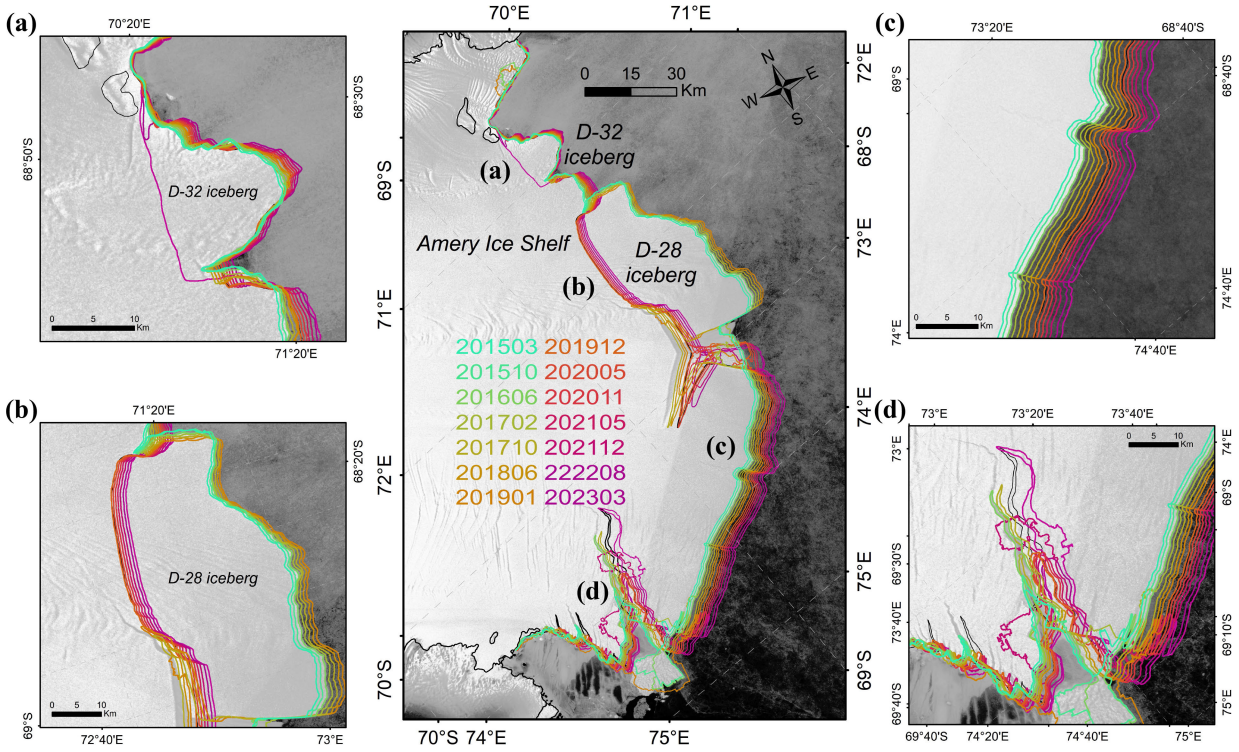


Fig. 6. Time-series front extraction results of the AIS. (a) and (b) Front near the D-32 and D-28 iceberg before and after they calved from AIS. (c) Gradual and stable expansion toward the Prydz bay. (d) Part of dramatically changing fronts of the AIS. The base map is the mosaic of Sentinel-1 imagery in EW mode in 2016.

delineation of the calving front is particularly challenging, primarily due to the potential misidentification of a substantial amount of floating sea ice and coastlines. Compared to the baseline method, our approach significantly enhances the detection accuracy of the Columbia Glacier front [see Fig. 5(e)]. The baseline can only identify one out of three parts of the calving front of the Columbia Glacier [26], while our method demonstrates superior performance, Transformer model.

In contrast to the conventional CNN baseline for calving front extraction frameworks, our model harnesses the synergy between the CNN and transformer architectures. The former, due to its inherent translational invariance and robust local feature extraction prowess, shows extensive utility in image processing tasks [67]. In contrast, the latter excels in spatial modeling in the context of long-range contextual dependencies. Integrating these two facets capitalizes on their respective strengths. Concurrently, the incorporation of a GLA mechanism is a pivotal catalyst in augmenting the network performance. By revealing and accentuating both local and global features and facilitating the interplay between the two, the network's latent potential is unlocked, channeling its focus toward discerning the distinctive features of glacier and ice shelf peripheries [23]. Finally, the cascaded pyramid structure inherent to the DeepLabv3+ network (ASPP) adeptly captures many multiscale, multifeature image insights [12] compared with the modified U-Net employed in the original CaFFe database [26]. The remarkable 45.6% improvement demonstrates the outstanding performance of our method, even in the more complex winter environment.

B. Ablation Study

To comprehensively explore and validate the performance of the Swin-transformer and global–local attention modules employed in our study, we conducted two additional sets of ablation experiments.

1) *Baseline Model:* DeepLabv3+ [25].
2) *Baseline Model + Swin-Transformer: STDeepLab [55]:* In the encoder layer of DeepLabv3+, we utilized the Swin-transformer for feature extraction.
3) *Proposed GLA-STDeepLab Method:* Introducing global–local attention into the Swin-transformer and integrating it into the DeepLabv3+ model.

The comparative analysis of the ablation experiments is presented in Table IV. In comparison to the baseline model, the incorporation of both the Swin-transformer and GLA modules demonstrates significant improvements in segmentation accuracy. Specifically, employing the Swin-transformer module alone yielded a 0.82% increase in precision and a 1.35% increase in IoU when compared to the baseline, which can be attributed to the Swin-transformer 's robust spatial feature extraction capabilities. Moreover, the integration of the global–local attention module into the Swin-transformer framework further elevated the segmentation accuracy. Notably, this integration resulted in a 1.83% improvement in precision and a 3.05% improvement in IoU. These exceptional outcomes of the ablation experiments validate the superior segmentation capabilities of our proposed model.

However, a crucial point of consideration lies in the integration of the Swin-transformer and GLA mechanisms into the network architecture. This integration leads to an increase

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TABLE IV
COMPARISONS OF THE NETWORK PERFORMANCE (THE DATA ARE BASED ON TEST DATASET (%); TRAINING SPEED REFERS TO THE AVERAGE TIME THE NETWORK TAKES PER EVERY EPOCH)

Scope	PA	Precision	Recall	F_1	IoU	Parameters (M)	Traing Speed (s)
Baseline (DeepLabv3+)	95.86	95.16	96.16	95.65	91.30	42.64	208
STDeepLab	96.43	95.98	96.67	96.32	92.65	28.60	278
GLA-STDeepLab	97.69	97.07	97.06	97.07	94.35	53.93	315

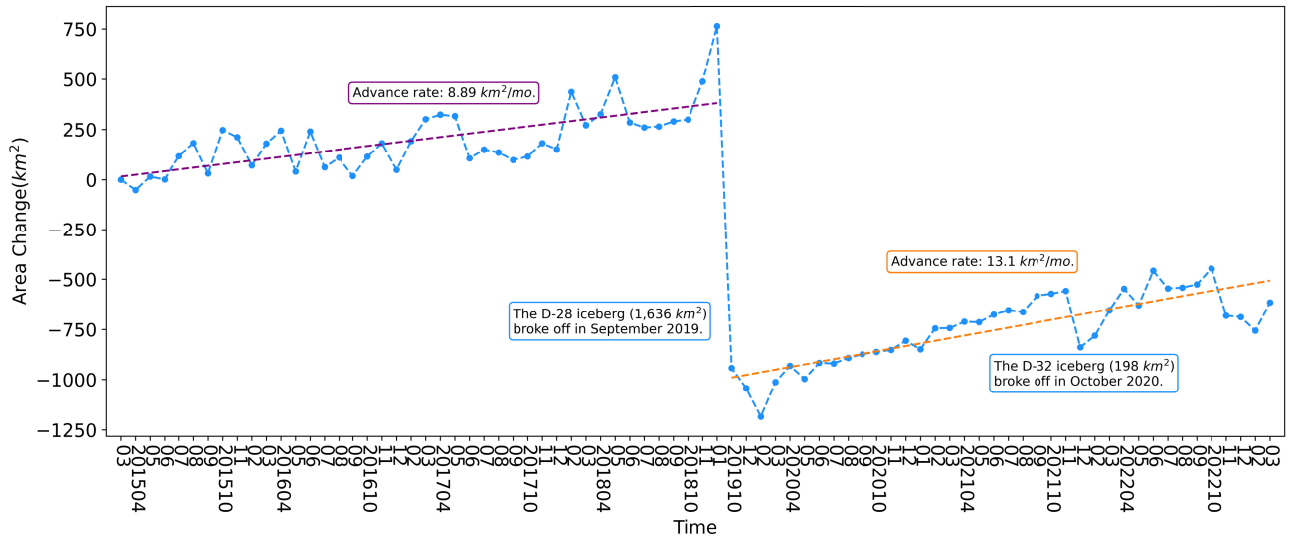


Fig. 7. Area changes of the AIS due to calving front fluctuations compared to March 2015. The lavender line indicates the area change trend of the AIS before the D-28 iceberg calving, and the orange–yellow line indicates after D-28 iceberg calving.

in the network's parameter count, reaching an impressive 53.93 million parameters. In contrast, the DeepLabv3+ network employing ResNet as its backbone boasts a more modest parameter count of 42.64 million, which is used in this study. Opting for a leaner backbone such as MobileNet [68] could further reduce the parameter count to a mere 11.25 million. In addition, the intricate computational processes intrinsic to the transformer architecture noticeably increase the training period, thereby amplifying the intricacies of network training [69]. Our empirical findings indicate that each training epoch of the GLA-STDeepLab network consumes 315 s, nearly double the temporal expenditure of training the benchmark DeepLabv3+ network.

We created virtual displays (Fig. 4) to showcase the glacier and ice shelf results extracted from the three models. The visual analysis revealed that the results obtained by GLA-STDeepLab exhibited a remarkable alignment with the ground truth, surpassing those of the baseline and STDeepLab models. Specifically, when extracting Jacobshavn Glacier, both the baseline and STDeepLab models erroneously misclassified certain areas of sea ice as glaciers due to the similar backscattering patterns caused by ice coverage. In contrast, the GLA-STDeepLab model effectively discriminated them by leveraging the fusion of global and local information.

C. Amery Ice Shelf Time Series

Leveraging the benefits of multiscale information extraction and global–local attention mechanisms within our network, we enable the simultaneous processing of diverse SAR data, thereby facilitating the extraction of valuable temporal information regarding glacier and ice shelf front changes. Temporal variations in the Amery Ice Shelf (AIS) front from March 2015 to March 2023 were analyzed using the methodology we proposed (see Fig. 6). The front detection algorithm yielded favorable validation results when applied to the AIS. We accurately captured the changes in the ice shelf coastline before and after the detachment of the D-28 iceberg, which had an approximate area of 1600 km^2 [see Fig. 6(b)]. Fig. 6(a) shows another glacier calving event that took place around October 2022. The D-32 iceberg, with an area of approximately 200 km^2, detached from the AIS and resulted in a loss of mass for the AIS. The precise detection of the D-28 and D-32 calving events provides compelling evidence for the accuracy and robustness of our methodology. Fig. 6(c) shows the gradual and stable expansion of the AIS from 2015 to 2023. In contrast, Fig. 6(d) highlights a significant variation in one section of the ice shelf front, where partial misclassifications occurred due to the complex ice shelf–ocean–sea ice environment in that particular area.

We conducted a comprehensive analysis of the area changes of the AIS relative to March 2015 (see Fig. 7). Prior to the calving event of the D-28 iceberg, the AIS exhibited a relatively modest expansion rate (8.89 km^2/mo.) with a slight but almost stable fluctuation. In September 2019, the detachment of the D-28 iceberg [70], [71], covering an area of approximately 1636 km^2, led to a significant reduction in the extent of AIS, confirming the influential role of calving

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in glacier mass loss. Subsequently, the AIS continued its expansion at a faster rate than that observed prior to the calving of D-28 (13.1 km^2/mo.), with a noticeable reduction in volatility from December 2019 to October 2020. Nevertheless, the overall extent of expansion remained significantly smaller when compared to the contribution of ice shelf calving. Ice shelf calving, along with basal melting, is the most important contributor to polar ice sheet mass loss and needs to be accurately monitored [72], [73].

In general, the methodology presented in this study offers a precise and effective means of accurately determining the positions of glacier and ice shelf fronts, making it of paramount importance for monitoring their development and facilitating comprehensive analysis. Furthermore, the monitoring of ice shelf calving has crucial significance in assessing variations in ice shelf and ice sheet mass, as well as advancing our understanding of their implications for sea-level rise and global climate change [74], [75].

VI. Conclusion

In this study, we present a novel network architecture, GLA-STDeepLab, designed for the detection of glacier and ice shelf fronts. We incorporate the Swin-transformer module into the DeepLabv3+ framework, significantly enhancing the modeling capability of long-range contextual dependencies in the CNN-based segmentation network. Moreover, by introducing GLA mechanisms into the Swin-TransDeepLab module, we effectively direct the network's focus toward capturing and expressing relevant information, thereby improving its segmentation performance. Through extensive experimentation on the comprehensive CaFFe database, our network achieves results with an IoU of 0.9435, an F1 score of 0.9707, and an MDE of 473 $\pm$ 34 m. Based on our approach, we successfully obtain precise time-series information for the AIS front during the period from 2015 to 2023 and demonstrate that the AIS exhibits stability with a gradual expansion, while recent mass loss primarily stems from the detachment of the D-28 and D-32 icebergs. The remarkable success of our model has greatly inspired us to further explore the applications of combining the transformer and CNN in the field and utilizing global–local attention mechanisms for remote sensing, particularly in polar regions, which will be the focal point of our future research endeavors.

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Qi Zhu received the bachelor's degree from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China. He is currently pursuing the Ph.D. degree in geographical information system with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

His research interests include the cryosphere remote sensing and computer vision. His research work focuses on the application of deep learning in polar regions.



Huadong Guo (Member, IEEE) is currently a Professor with the Aerospace Information Research Institute and an Academician with the Chinese Academy of Sciences (CAS), Beijing, China. He has published more than 510 papers and 24 books. He specializes in remote sensing, radar for Earth observation, and digital Earth science.

Mr. Guo is a fellow of the World Academy of Sciences (TWAS), a Foreign Member of the Russian Academy of Sciences, and a Foreign Member of the Finnish Society of Sciences and Letters. He served as a member for the UN 10-Member Group to support the Technology Facilitation Mechanism for SDGs from 2018 to 2021 and the Chairperson for the International Committee of Remote Sensing of Environment from 2017 to 2020. He is the awardee of 18 domestic and international prizes. He serves as the Honorary President for the International Society for Digital Earth (ISDE), the Director for the International Center on Space Technologies for Natural and Cultural Heritage under the Auspices of UNESCO, and the Chair for the Digital Belt and Road Program (DBAR).



Lu Zhang received the Ph.D. degree in synthetic aperture radar (SAR) remote sensing from the Institute of Remote Sensing Application, Chinese Academy of Sciences (CAS), Beijing, China, in 2008.

He was a Visiting Scholar at the Institute of Electronics and Telecommunications of Rennes, University of Rennes 1, Rennes, France, in 2013. He is currently an Associate Professor with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, CAS. His research interests include SAR image processing, information extraction, physical parameter estimation models, big Earth data, moon-based Earth observation methods, and polar remote sensing. He is the author of more than 90 journal articles and nine patents in these fields.



Dong Liang received the B.Sc. degree in applied mathematics from the University of Hull, Hull, U.K., in 2006, and the M.Sc. degree in applied mathematics from Mälardalens University, Västerås, Sweden, in 2009. He is currently pursuing the Ph.D. degree in cartography and geoscience information system (GIS) with the University of Chinese Academy of Sciences, Beijing, China, with a focus on big Earth data and polar remote sensing.

Mr. Liang serves as the Deputy Secretary-General for the Chinese National Committee of the International Society for Digital Earth.



Zherong Wu (Member, IEEE) received the B.Eng. degree in remote sensing from Wuhan University, Wuhan, China, in 2019, and the Ph.D. degree in Earth system and geoinformation science from The Chinese University of Hong Kong, Hong Kong, in 2023.

She was a Research Intern at the United Nations, San Francisco, CA, USA, in 2020. She is currently a Post-Doctoral Fellow with the College of Agriculture and Life Sciences, Cornell University, Ithaca, NY, USA. Her research interests focus on satellite remote sensing and deep learning for sustainable development goals.



Yiming Liu received the bachelor's degree from the School of Earth and Space Sciences, Peking University, Beijing, China, where he is currently pursuing the Ph.D. degree in photogrammetry and remote sensing.

His research interests include correlation analysis and big Earth data.



Zhuoran Lv received the bachelor's degree in geography from the China University of Geosciences, Wuhan, China, in 2021. She is currently pursuing the Ph.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

Her main research interests include remote sensing and sustainable development. Her research work focuses on the application of nighttime light in urban regions.

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